

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1180

Hubbardstown 138 kV
May 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1180. The Interconnection Customer proposes a 84.6 MW solar photovoltaic facility interconnecting at the Hubbardstown 138 kV substation in the Kentucky area. The request was received on April 02, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 438 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,111,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MACOMB - BARTLETTS FERRY 138 kV
Contingency: VALEMOUNT - TRISTATE 345 kV line
Loading: 114.7% (Rating: 261 MVA)
DFAX: 5.8%
Pre-project loading: 112.8%
2. Facility: HEWETT - MACOMB SUBSTATION 138 kV L
Contingency: VALEMOUNT - TRISTATE SUBSTATION 345 kV
Loading: 114.5% (Rating: 260 MVA)
DFAX: 5.8%
Pre-project loading: 112.6%
3. Facility: DAVIS RD - HAGAN 138 kV
Contingency: HUBBARDSTOWN SUBSTATION - POPLAR BLUFF GENERATING STA 345 kV line
Loading: 113.8% (Rating: 277 MVA)
DFAX: 9.2%
Pre-project loading: 111.0%
4. Facility: HAGAN - SIGGN 138 kV L
Contingency: HUBBARDSTOWN - POPLAR BLUFF GENERATING STA 345 kV L
Loading: 113.6% (Rating: 279 MVA)
DFAX: 9.2%
Pre-project loading: 110.8%
5. Facility: TEORO - BLAND SUBSTATION 138 kV
Contingency: VALEMOUNT - TRISTATE SUBSTATION 345 kV
Loading: 109.5% (Rating: 278 MVA)
DFAX: 6.7%
Pre-project loading: 107.5%

6. Facility: TESORO - BOWLES 138 kV
 Contingency: VALEMOUNT - TRISTATE 345 kV
 Loading: 108.5% (Rating: 292 MVA)
 DFAX: 6.7%
 Pre-project loading: 106.5%
7. Facility: FIG GARDEN - BOWLES 138 kV line
 Contingency: VALEMOUNT - TRISTATE 345 kV
 Loading: 108.3% (Rating: 295 MVA)
 DFAX: 6.7%
 Pre-project loading: 106.3%
8. Facility: SIGGEN - MARTINEZ SUBSTATION 138 kV
 Contingency: HUBBARDSTOWN - TRISTATE 345 kV
 Loading: 100.5% (Rating: 264 MVA)
 DFAX: 11.9%
 Pre-project loading: 96.7%
9. Facility: VALEMOUNT 138/1 kV Transformer
 Contingency: PENCE SUBSTATION - VALEMOUNT 345 kV
 Loading: 100.4% (Rating: 543 MVA)
 DFAX: 6.2%
 Pre-project loading: 99.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MACOMB - BARTLETTS FERRY 138 kV	Rebuild line with higher-capacity conductor	\$15,819,000	100% (\$15,819,000)
HEWETT - MACOMB SUBSTATION 138 kV L	Rebuild line with higher-capacity conductor	\$9,728,000	100% (\$9,728,000)
DAVIS RD - HAGAN 138 kV	Rebuild line with higher-capacity conductor	\$11,002,000	66% (\$7,273,000)
HAGAN - SIGGN 138 kV L	Rebuild line with higher-capacity conductor	\$3,505,000	66% (\$2,317,000)
TEORO - BLAND SUBSTATION 138 kV	Reconductor line	\$1,402,000	42% (\$588,000)
TESORO - BOWES 138 kV	Reconductor line	\$2,898,000	42% (\$1,215,000)
FIG GARDEN - BOWLES 138 kV line	Reconductor line	\$2,071,000	42% (\$869,000)
SIGGEN - MARTINEZ SUBSTATION 138 kV	Reconductor line	\$1,477,000	9% (\$138,000)
VALEMOUNT 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$6,278,000	3% (\$164,000)
	Total allocated to project		\$38,111,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.